

# MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)

## PLANT TECHNICAL AUDIT & VIBRATION INSPECTION REPORT

|              |                                   |                  |                               |
|--------------|-----------------------------------|------------------|-------------------------------|
| Report No:   | MRPL/INSP/2026/CDU-042            | Inspection Date: | 31-Aug-2026                   |
| Plant Unit:  | Crude Distillation Unit-1 (CDU-1) | Lead Inspector:  | R. K. Sharma (Sr. Insp. Eng.) |
| Asset Group: | Primary Hydrocarbon Pumps         | Status:          | ACTION REQUIRED - SEVERE      |

### 1. Executive Summary

During routine thermographic and vibration screening in CDU-1 on 31-Aug-2026, severe abnormal vibration and elevated drive-end bearing temperatures were detected on Main Crude Charge Pump **P-102A**. Spectral analysis indicates progressive bearing inner-race spalling and impending mechanical seal failure. Immediate controlled shutdown and overhaul are required to prevent catastrophic hydrocarbon leakage.

### 2. Equipment Inspection Findings

| Tag    | Equipment Name                | Observed Parameter    | Recorded  | Threshold | Severity | Action             |
|--------|-------------------------------|-----------------------|-----------|-----------|----------|--------------------|
| P-102A | Crude Charge Pump A           | DE Bearing Vibration  | 7.8 mm/s  | 4.5 mm/s  | CRITICAL | Immediate Overhaul |
| P-102A | Crude Charge Pump A           | DE Bearing Temp       | 98.5 °C   | 75.0 °C   | CRITICAL | Replace Bearing    |
| P-102B | Crude Charge Pump B (Standby) | DE Bearing Vibration  | 2.1 mm/s  | 4.5 mm/s  | NORMAL   | Switch to Lead     |
| V-104  | Atmospheric Column Feed Valve | Stem Leakage Rate     | 0.02 L/hr | 0.01 L/hr | WARNING  | Re-pack Gland      |
| E-101  | Crude Pre-heat Exchanger      | Differential Pressure | 1.4 bar   | 1.8 bar   | NORMAL   | Continue Monitor   |

### 3. Recommendations & Corrective Actions

1. Immediately transfer crude distillation load from P-102A to standby pump P-102B.
2. Isolate, depressurize, and de-inventory pump P-102A following MRPL SOP-CDU-SEC-014.
3. Dismantle DE bearing housing, replace SKF 23144 CC/W33 bearing assembly, and install new John Crane Type 5620 mechanical seal.
4. Estimated overhaul cost: **INR 2,85,000** (Spares: INR 1,95,000; Manpower/Tooling: INR 90,000).
5. Estimated downtime: 18 hours with zero throughput loss due to P-102B readiness.